CHAPTER

3

THERMAL EXPANSION AND PIPING FLEXIBILITY

The most basic requirement of piping stress analysis is to ensure adequate flexibility in the piping system for absorbing the thermal expansion of the pipe. Prior to the nuclear power era, almost all discussions and treatments of pipe stress were concentrated in piping flexibility analysis. The works by Spielvogel [1], Kellogg [2], Olson and Cramer [3], and Brock [4] are some of these earlier examples. Even nowadays, despite the fact that stress analysis covers much more than flexibility analysis, engineers still tend to regard pipe stress analysis as just a fancy term for pipe flexibility analysis. In any case, piping flexibility remains as one of the most important tasks in piping engineering and stress analysis.

3.1 THERMAL EXPANSION FORCE AND STRESS

The pipe expands or contracts due to temperature changes in the pipe. In this discussion, the term expansion implies either expansion or contraction. When a pipe expands it causes the entire piping system to make room for its movement. This creates forces and stresses in the pipe and on its connecting equipment. If the piping system does not have enough flexibility to absorb this expansion, the force and stress generated can be large enough to damage the piping and the connecting equipment. An idealized straight pipe as shown in Fig. 3.1(a) will be used to investigate the potential magnitude of the force and stress that can be generated [5].

3.1.1 Ideal Anchor Evaluation

Figure 3.1(a)(1) shows a straight pipe connected to two ideal anchors. An ideal anchor has infinite stiffness such that no anchor movement is generated regardless of the magnitude of the force applied. The anchors implemented in computer programs are mostly ideal anchors. In our example, when the temperature of this two-anchored straight pipe changes, it causes the pipe to expand. However, the anchors at the ends prevent it from expanding. The resistance of the anchor generates force on the anchors from which the same force is reflected back to the pipe. Figure 3.1(a)(2) shows that when one end of the pipe is loose, the pipe has a free expansion equal to $\Delta = \alpha L(T_2 - T_1)$, where α is the thermal expansion rate of the pipe and $(T_2 - T_1)$ is temperature change. There is no force or stress generated in the free expansion state. However, because neither end is loose in this two-anchored case, the force generated on the anchor is equivalent to the force required to push the free expanded end back to its original position. The squeezing movement is equal to the strain times the length. That is,

$$\Delta = \alpha(T_2 - T_1)L = eL = \frac{S}{E}L = \frac{F}{EA}L$$

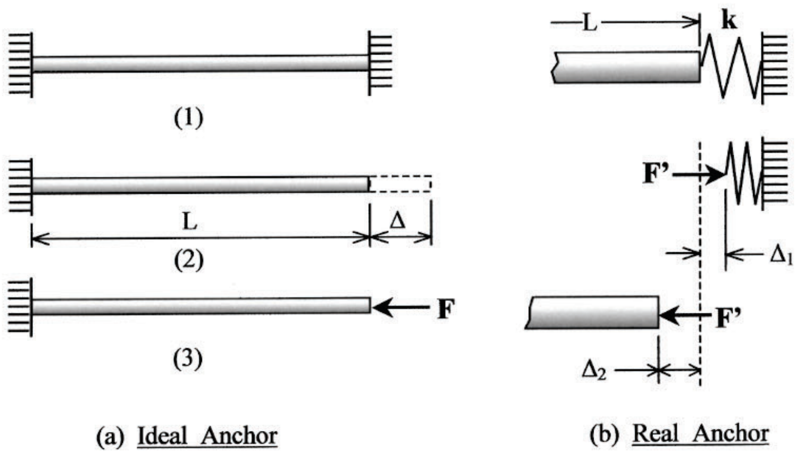


FIG. 3.1
THERMAL EXPANSION FORCE

or

$$F = EA\alpha(T_2 - T_1), \quad S = E\alpha(T_2 - T_1) \tag{3.1}$$

where E is the modulus of elasticity of the pipe material, A is the cross-section area of the pipe, F is the anchor force, and S is the axial stress. With ideal anchors, the force and stress generated are independent of the length of the pipe. The force and stress generated are huge even for a small pipe section with only a moderate temperature change. For example, increasing the temperature of a 6-in. standard wall carbon-steel pipe from 70°F ambient condition to 300°F operating condition creates an axial stress of 45,000 pounds per square inch (psi) and an axial force of 250,000 pounds (lbs) in the pipe and on the anchors.

The force and the stress are excessive even though the temperature is only 300°F. This shows that a straight-line direct piping layout is generally not acceptable. The ideal anchor simulation also shows that the magnitudes of the forces and the stress are the same regardless of pipe length. This idealized conclusion sometimes becomes a handicap when designing a closely spaced and tight piping system.

3.1.2 The Real Anchor

Equation (3.1) shows the potential magnitudes of the force and the corresponding stress that can be generated by a temperature change in the pipe. These quantities are independent of the length of the pipe. This is the conclusion that engineers normally obtain from a theoretical treatment on the subject. In the real world, however, the stiffness of the anchor is limited. A stiffness of 10^6 lb/in. is already very difficult to achieve due to the flexibility of the structure, foundation, and attachments themselves. Depending on pipe size, the practical anchor stiffness more likely ranges from 10^4 to 10^7 lb/in.

Figure 3.1(b) shows an interaction of a real anchor with an expanding straight pipe. Here, only one end of the pipe is connected to a real anchor with a stiffness k. The other end is still connected to an ideal anchor of infinite stiffness. Because of its flexibility, the real anchor will absorb part of the pipe expansion, Δ_1, and the pipe itself absorbs the rest of the expansion, Δ_2. The combination of Δ_1 and Δ_2 is the total expansion, Δ. Because both anchor and pipe receive the same force, F', we have

$$\Delta_1 = \frac{F'}{k}, \quad \Delta_2 = \frac{F'L}{EA}$$

$$\Delta = \alpha L(T_2 - T_1) = \Delta_1 + \Delta_2 = \frac{F'}{k} + \frac{F'L}{EA} = F'\left(\frac{1}{k} + \frac{L}{EA}\right)$$

or

$$F' = \frac{\alpha L(T_2 - T_1)}{L\left(\frac{1}{kL} + \frac{1}{EA}\right)} = \frac{\alpha(T_2 - T_1)EA}{\left(\frac{EA}{kL} + 1\right)} \tag{3.2}$$

Equation (3.2) shows that with a real anchor, the force generated is dependent on the length and anchor stiffness. For short pipes, the force is roughly proportional to the length of the pipe. The equation includes an additional term, EA/kL. Another equivalent term would have been likewise included if both ends of the pipe were connected to real anchors. This term is important when dealing with short pipes. Take the 6-in. pipe discussed previously, for example; a 5-ft pipe with one end connected to an anchor having a stiffness of 10^5 lb/in. will result in a force of 8800 lb and a stress of 1600 psi. When both ends are connected to real anchors, the force is further reduced to 4500 lb and the stress to 800 psi. These numbers are roughly just 1/30 and 1/60, respectively, of the numbers obtained by ideal anchors.

From the above, it is clear that the stiffness of the anchor has a very great effect on the forces and stresses generated. This is especially true for close layouts at moderate temperature ranges. In an actual analysis, it is important to obtain as accurate as possible the stiffness of the anchors and restraints. However, obtaining accurate anchor and restraint stiffness is a very complicated process involving the structural analysis of the support structure, foundation, and attachment. It is not feasible for a routine piping stress analysis to include the accurate anchor and restraint stiffness. Generally, the ideal anchor and restraint are used in conjunction with engineering judgment.

3.2 METHODS OF PROVIDING FLEXIBILITY

There are two main categories of methods for providing piping flexibility: the flexible joint method and the pipe loop method. Flexible joints, including expansion joints, ball joints, and others, are discussed in Chapter 7. This chapter discusses the method of using pipe loops and offsets to provide flexibility.

From Fig. 3.1 we know that the huge thermal expansion force and stress on an anchored straight section of pipe are the result of squeezing the free expansion axially back to the pipe. This is very difficult, as we can experience by squeezing the ends of a wooden stick. Instead of this direct squeezing, we can absorb the same amount of movement much easier by bending the stick sideways. This is the principle of providing piping flexibility. The flexibility is provided by adding a portion of the piping that runs in the direction perpendicular to the straight line connecting two terminal fixation points. Figure 3.2 shows an expansion loop used in a long straight pipe run. With the loop, the pipe expands into the loop by bending the legs of the loop instead of squeezing the pipe axially. The longer the loop leg, the lesser the force generated in absorbing a given expansion. From the basic beam formula given in Table 2.1, we know that the required force is inversely proportional to the cube of the leg length, and the generated stress is inversely proportional to the square of the leg length. A small increase in loop leg length has a considerable reduction effect on force and stress.

3.2.1 Estimating Leg Length Required

The required leg length can be estimated via the guided cantilever approach. The method is explained by using the L-bend given in Fig. 3.3 as an example.

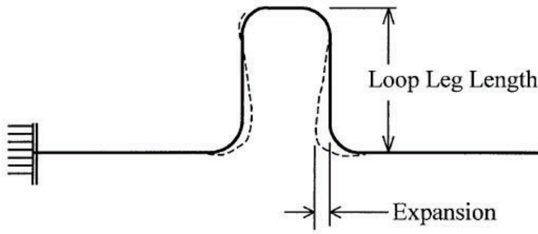


FIG. 3.2
PIPE EXPANSION LOOP

When the piping system is not constrained and is free to expand as in Fig. 3.3(a), points B and C will move to B' and C', respectively, due to thermal expansion. The end point C moves Δ_x and Δ_y amounts in x and y directions, respectively, but no internal force or stress is generated in the absence of a constraint. However, in the actual case, the ends of the piping are always constrained as in Fig. 3.3(b). This is equivalent to moving the free expanded end C' back to the original point C, and forcing point B to move to point B''.

The deformation of each leg can be assumed to follow the guided cantilever shape shown in Fig. 2.14(b). From a flexibility point of view, this is conservative because the end rotation is ignored. The force and stress of each leg can now be estimated by the guided cantilever formula. For this simple L shape, leg AB is a guided cantilever subject to Δ_y displacement, and leg CB is a guided cantilever subject to Δ_x displacement. The stress at each leg is mainly the beam bending stress caused by the expansion displacement. From the cantilever beam formula, we can estimate the stress at each leg as follows

$$S = \frac{M}{Z} = \frac{1}{Z}\frac{6EI}{L^2}\Delta = \frac{1}{\pi r^2 t}\frac{6E\pi r^3 t}{L^2}\Delta = \frac{6Er}{L^2}\Delta = \frac{3ED}{L^2}\Delta \tag{3.3}$$

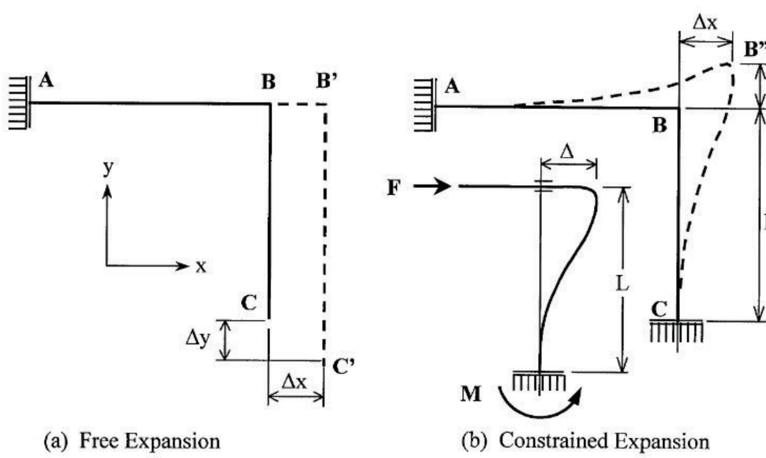


FIG. 3.3
EXPANSION STRESS BY GUIDED CANTILEVER APPROACH

The approximate formulas $Z = \pi r^2 t$ and $I = \pi r^3 t$ are used, respectively, for the section modulus and moment of inertia of the pipe cross-section. Equation (3.3) is a convenient formula for quick estimation of the expansion stress. By substituting $E = 29.0 \times 10^6$ psi and $S = 20{,}000$ psi, Eq. (3.3) reduces to Eq. (3.4a) for finding the leg length required for steel pipes

$$L = \sqrt{\frac{3ED}{S}\Delta} = 66\sqrt{D\Delta} \tag{3.4a}$$

Equation (3.4a), although derived from psi units of E and S, is applicable to all consistent units. L, D, and Δ can be expressed in inches or in millimeters.

3.2.2 Inherent Flexibility

Piping in a plant generally runs through a few turns before connecting to a fixation point such as a vessel or rotating equipment. These turns and offsets generally provide enough flexibility to absorb the expansion displacement without causing excessive stress in the pipe. The American Society of Mechanical Engineers (ASME) B31 piping code [6] has provided a criterion as a measure of adequate flexibility, subject to other requirements of the code. The code states that no formal thermal expansion flexibility analysis is required when "The piping system is of uniform size, has not more than two anchors and no intermediate restraints, is designed for essentially non-cyclic service (less than 7000 total cycles), and satisfy the following approximate criterion:"

(1) English units

$$\frac{DY}{(L-U)^2} \leq 0.03 \tag{3.4b}$$

(2) SI units

$$\frac{DY}{(L-U)^2} \leq 208.3 \tag{3.4c}$$

Equations (3.4b) and (3.4c) are in conventional units

where

D = nominal pipe size, in (mm)
Y = resultant of movement to be absorbed by piping system, in (mm)
L = developed length of piping system between two anchors, ft (m)
U = anchor distance (length of straight line joining anchors), ft (m)

Equations (3.4a) and (3.4b) are practically equivalent when consistent units are used. If the surplus length $(L - U)$ is considered as the leg length perpendicular to the line of expansion, Eq. (3.4b) can be converted to Eq. (3.4a) with a constant of 69 instead of 66.

3.2.3 Caution Regarding Quick Check Formulas

During the era when flexibility analysis of a rather simple system could take a couple of weeks of hard work by a specialist engineer to accomplish, quick formulas such as Eq. (3.4a) would mean the difference between whether a plant could be constructed on schedule. These formulas have been extensively taught in various training classes. However, in this age of high-speed computers, these formulas have only very limited use. They might be used by field specialists when surveying a plant for problem installations, or occasionally by design engineers at a remote site. However, at an operating plant or an engineering office, the analysis is better performed by using a quick and accurate computer program. An accurate analysis of a fairly complicated piping system takes only an hour or so to prepare the data and to run with a computer program.

By limiting the pipe stress to about 20,000 psi, Eqs. (3.4a), (3.4b) and (3.4c) are adequate for protecting the piping itself. However, because piping is always connected to certain equipment, this 20,000-psi stress will most likely generate too much load for the equipment to take. Moreover, as will be discussed later in this chapter, some piping components are associated with stress intensifications that are not included in the equations — although a component, such as a bend, that has significant stress intensification may also possess significantly added flexibility. This mutual compensation of stress intensification and the added flexibility validates the usefulness of the equations. However, it is worth noting that not all stress intensifications come with added flexibility.

3.2.4 Wall Thickness and Thermal Expansion Stress

Because expansion stress is calculated by dividing the moment, M, with the section modulus, Z, engineers might be wrongly tempted to increase the wall thickness to reduce the expansion stress. An increase in wall thickness increases the section modulus, but also proportionally increases the moment of inertia. The section modulus is defined as $Z = I/r_o$, which is directly proportional to the moment of inertia. Therefore, the first consequence of increasing the wall thickness is an increase in bending moment under a given thermal expansion. This increased moment divided by the proportionally increased section modulus ends up with the same stress as before, prior to the increase of the thickness. The thicker wall thickness does not reduce the thermal expansion stress. It only unfavorably increases the forces and moments in the pipe and at the connecting equipment. Therefore, as far as thermal expansion is concerned, the thinner the wall thickness the better it will be for the system.

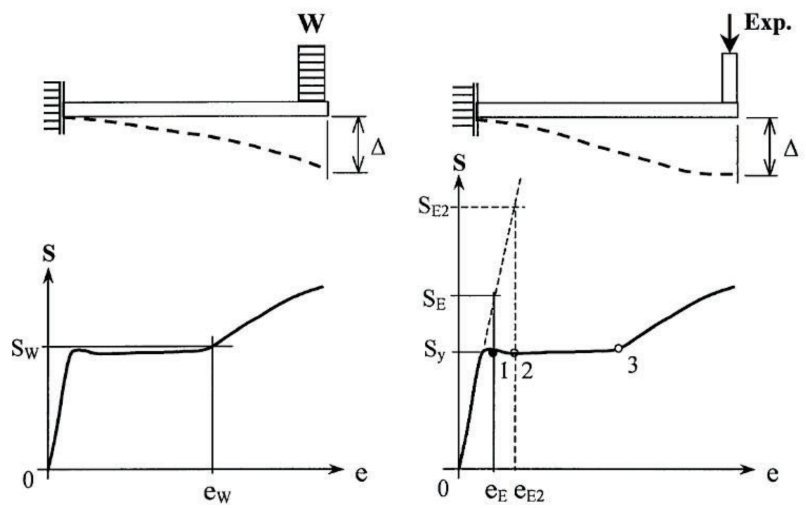
3.3 SELF-LIMITING STRESS

The stress generated by thermal expansion is self-limiting in nature. This self-limiting stress behaves quite differently from the sustained stress caused by weight and pressure. Figure 3.4 shows the differences between these two types of stresses.

Figure 3.4(a) is a pipe subject to weight load, thus generating sustained stress in the pipe. By increasing the weight gradually, the stress and the accompanying displacement also increase accordingly. When the stress reaches the yield point of the material, the stress maintains the same magnitude, but the displacement increases abruptly by a large amount. (This is an oversimplification, because the bending stress does not exactly behave this way.) In sustained loading, the stress generated has to be in static balance with the applied load — that is, stress is always proportional to the load, but displacement depends on the characteristics of the material. In this case, if the stress, S_w, exceeds the yield strength slightly, a large strain, e_w, will be created. This strain is so large that it results in a gross deformation in the system. Therefore, sustained stress is generally limited to values lower than the yield strength of the material.

Figure 3.4(b) shows the case for self-limiting stress. When the pipe is subject to thermal expansion or other displacement load, the mechanism of balance shifts to the strain — that is, the strain always corresponds with the amount of expansion or displacement. Once the displacement reaches its potential magnitude, the whole thing stops because the displacement has only that much. The generated stress depends on the characteristics of the material. For instance, when the strain corresponding to the displacement exceeds the yield strain, it stays there without any further abrupt movement. This position fixing nature is called self-limiting. The stress due to thermal expansion is self-limiting stress.

Self-limiting stress has the following characteristics: (1) Generally, it does not break ductile pipe in one application of the load. (2) Its mode of failure is fatigue requiring many cycles of applications. (3) Fatigue failure depends on stress range, measured from the lowest stress to the highest stress in the



(a) Sustained Stress

(b) Self-Limiting Stress

FIG. 3.4
SUSTAINED VS. SELF-LIMITING STRESSES

operating cycles. (4) Stress can reach yield strength without causing failure or gross system deformation. (5) When dealing with stresses at or beyond yield strength, the failure is evaluated by the strain range rather than the stress range.

3.3.1 Elastic Equivalent Stress

The actual stress of a self-limiting stress rarely exceeds the yield strength. As shown in Fig. 3.4(b), even if the self-limiting strain exceeds the yield strain, the actual stress is still equal to the yield strength. This nature makes it difficult to evaluate the effect of self-limiting stress using the actual stress. For instance, points 1, 2, and 3 all correspond to the same stress, yet we know point 3 is much more critical than the others. Therefore, it is natural that self-limiting stress should be evaluated by the strain rather than by the stress. However, because the analyses are traditionally set up to calculate stress and they are also done by elastic methods, the determination of the actual strain is very complicated. To overcome this shortcoming, the elastic equivalent stress is used to measure the strain. The elastic equivalent stress is simply the product of the strain and modulus of elasticity. This is equivalent to replacing the actual stress-strain curve with a straight line extending from the initial straight portion of the curve — that is, the elastic equivalent stress for strain e_E is S_E, and for e_{E2} it is S_{E2}, etc. This elastic equivalent stress can be several times higher than the yield stress, yet the actual stress is still the same as the yield stress.

3.4 STRESS INTENSIFICATION AND FLEXIBILITY FACTORS

A piping system consists of many different components such as bends, elbows, reducers, tees, valves, and flanges. However, in the analysis we normally idealize these various components into two types of elements: the straight pipe beam element and the curved pipe beam element.

3.4.1 Ovalization of Curved Pipes

A piping system depends mainly on its bending flexure to absorb thermal expansion and other displacement loads. When a straight pipe is subject to bending, it behaves like any straight beam: its cross-section remains circular and the maximum stress occurs at the extreme outer fiber. However, under a bending moment, a curved pipe element behaves differently from that of a solid curved beam. When subject to a bending moment, the circular cross-section of the bend becomes oval. This is the famous ovalization we are all aware of. Figure 3.5 shows the ovalization associated with in-plane bending. An out-of-plane bending, on the other hand, produces an oblique ovalization inclining at an angle with the major axes. The ovalization tendency of the curved pipe has resulted in the following peculiar phenomena:

(1) Increase of flexibility. Ovalization is caused by the relaxation of the extreme outer fiber of the bend. Without the proper participation of the extreme outer fiber, the effective moment of inertia of the cross-section is reduced. This reduction in effective moment of inertia increases the flexibility of the bend over the non-ovalized theoretical bend by a factor [6] of

$$k = 1.65/h \tag{3.5}$$

where k is the flexibility factor and $h = tR/r_m^2$ is the bend flexibility characteristic.

(2) Increase of longitudinal bending stress. The relaxation of the extreme outer fiber has shifted the maximum longitudinal stress due to bending to a location away from the extreme outer fiber location. This reduction in the moment resisting arm of the high stress portion is equivalent to reducing the effective section modulus of the cross-section. The maximum longitudinal stress is, therefore, greater than the maximum stress obtained by the elementary bending theory. The ratio of the two stresses is the stress intensification factor (SIF). That is,

$$i = \frac{S}{M/Z} \tag{3.6}$$

The theoretical longitudinal SIFs are related to the bend flexibility characteristic as [2]

$$i_{Li} = \frac{0.84}{h^{2/3}} \quad \text{for in-plane bending} \tag{3.7}$$

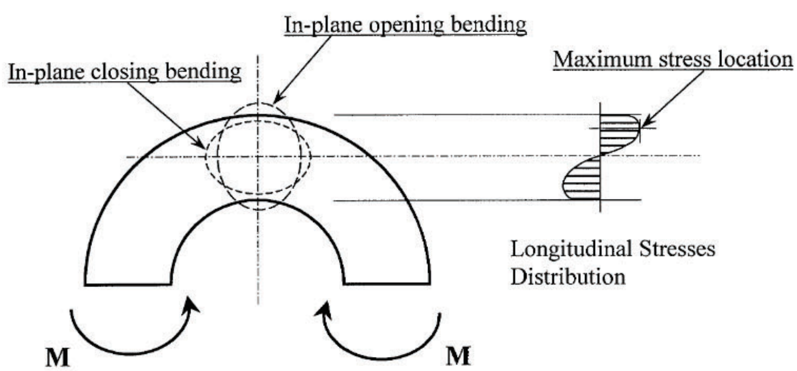


FIG. 3.5
OVALIZATION OF BEND UNDER EXTERNAL BENDING

$$i_{Lo} = \frac{1.08}{h^{2/3}} \quad \text{for out-of-plane bending} \tag{3.8}$$

(3) Creation of circumferential shell bending stress. Squeezing the circular cross-section into an oval shape generates bending on the pipe wall. This, in turn, creates a high circumferential bending stress on the pipe wall. Because this shell bending stress is non-existent on a circular cross-section, there is no direct comparison with the non-ovalized bend. For the sake of convenience, the stress is compared with the flexure bending stress of a circular cross-section as shown in Eq. (3.6).

The theoretical SIFs for the circumferential stresses are [2]

$$i_{Ci} = \frac{1.80}{h^{2/3}} \quad \text{for in-plane bending} \tag{3.9}$$

$$i_{Co} = \frac{1.50}{h^{2/3}} \quad \text{for out-of-plane bending} \tag{3.10}$$

The flexibility factor given in Eq. (3.5) is used by ASME codes for both in-plane and out-of-plane bending. The theoretical SIFs given by Eqs. (3.7), (3.8), (3.9), and (3.10) are used only in Class 1 nuclear piping [7]. For other ASME piping codes, only one-half of the theoretical values are used instead.

3.4.2 Code SIFs

Earlier piping stress analyses were mainly concerned with the flexibility of piping subject to thermal expansion. As previously discussed, the failure mode of self-limiting expansion stress is fatigue due to repeated operations. Therefore, to validate these SIFs, the most direct and logical approach is the fatigue test. After many tests and researches, Markl and others [8–10] have found that theoretical SIFs are consistent with the test data. However, tests performed on commercial pipe also revealed an SIF of almost 2.0 against a polished homogeneous tube with regard to fatigue failure. This factor is mainly due to the unpolished weld effect, or clamping effect at fixing points, combined with the less than perfectly homogeneous commercial pipe. To simplify the analysis procedure, the applicable SIFs are taken based on the commercial girth welded pipe as unity. This, in effect, reduces the applicable SIFs to just one-half of the theoretical SIFs given by Eqs. (3.9) an (3.10). That is, for bends we have:

- In-plane bending SIF

$$i_i = \frac{0.90}{h^{2/3}} \tag{3.11}$$

- Out-plane bending SIF

$$i_o = \frac{0.75}{h^{2/3}} \tag{3.12}$$

The preceding equations are for bends. For other components, Markl [10] has succeeded in using equivalent bends as shown in Fig. 3.6 [12] to arrive at SIFs that are comparable with the test results. Using the equivalent elbows and making adjustments for actual crotch radius and thickness, a set of SIFs [11] for various components was constructed with a single flexibility characteristic parameter, h. The SIF for a welding tee, for example, can also be expressed by Eq. (3.11) by setting the flexibility characteristic, $h = 4.4t/r$ (recently revised to $h = 3.1t/r$), from the equivalent elbow characteristics. However, in contrast to smooth bends, the SIF for out-of-plane bending is generally greater than that for in-plane bending in miter bends, welding tees, and other branch connections.

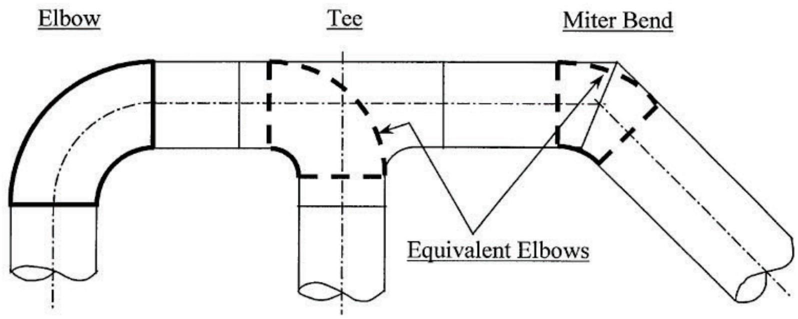


FIG. 3.6
EQUIVALENT ELBOWS

Although they remain mostly unchanged, these SIFs have nevertheless been continuously revised through the years. The values given in the current edition of the applicable code should always be used.

Stress intensification factors. SIFs are only one-half of the theoretical factors, and are intended for use only on self-limiting stresses. By using the commercial pipe with an unpolished girth weld as the basis, the code SIFs as given by Eqs. (3.11) and (3.12) are only one-half of the theoretical SIFs. The adoption of this basis is mainly attributed to practicality. If the theoretical SIF were used, then an analysis would have to identify all girth weld locations for applying the SIF. This is not very practical when large amounts of piping components are involved. Currently, only Class 1 nuclear piping uses the theoretical SIF.

When an SIF is involved, the stress calculated using the ASME B31 code formula is only one-half of the theoretical stress. This does not cause problems if everything is done within the range specified by the code, because the allowable stress has also been adjusted accordingly. However, there are occasions when something outside the code has to be referenced. For instance, when dealing with steady-state vibrations, pipe stress has to be evaluated with a fatigue curve that is generally constructed with theoretical stresses. In this case, the stress calculated by the B31 code has to be doubled before being applied to the fatigue curve.

By comparing Eqs. (3.11) and (3.12) with Eqs. (3.9) and (3.10), it is clear that the code SIF is the measure of the circumferential stress. Failure locations on specimens used in fatigue tests also showed that the SIF is due to the circumferential stress. Because circumferential stress is a shell bending stress that does not provide any static equilibrium to the load applied, it has little significance in the sustained load. Therefore, the code SIFs derived from fatigue tests and theoretical circumferential stresses are only applicable to self-limiting loads that produce fatigue in the pipe.

Sustained loads. For sustained loads, a separate set of SIFs is required. This separate set of SIFs for the sustained load has been used in Class 2 and Class 3 nuclear piping [7]. However, for non-nuclear piping systems, a separate set of SIFs is not provided for sustained loads. To this end, there are several practices used in various industries to deal with this matter. One of these practices also uses the same code SIF for sustained loads. This is a conservative approach used by non-discriminating engineers. Another practice completely ignores the SIF for sustained loads. This is suggested mainly by engineers who have been involved in the earlier development of SIFs. The rationale is that code SIFs are for fatigue only. However, it is recognized that some type of SIF is needed for sustained loading. One approach is to use the same set of SIFs intended for self-limiting loads (code SIFs) applied with a constant modification factor, which is somewhat less than 1.0. This approach, although not accurate, is in the proper practical range.

The SIF for sustained load is more closely related to the load-resisting longitudinal stresses given by Eqs. (3.7) and (3.8). These longitudinal stresses have the same mathematical format as the circumferential stresses given in Eqs. (3.9) and (3.10), except that their relative strengths are switched. On circumferential stresses, in-plane bending produces greater stress, whereas with longitudinal stresses, out-of-plane bending produces a larger stress. Therefore, theoretically, it is not possible to use a constant modification factor to relate these two sets of stresses. However, if only the greater stress intensifications on each set are used, then it is possible to conservatively use a constant modification factor. This is the approach used by ASME B31.1 [6], which uses a modification factor of 0.75. For other ASME B31 codes that use different in-plane and out-of-plane SIFs, the constant modification factor approach may not be suitable.

Tables for flexibility factors and SIFs. Each code has a table or tables that list and explain the flexibility factors and SIFs for most of the common components used in a piping system. Although flexibility factors, used to calculate the piping forces and moments, are all similar, SIFs differ slightly between the codes. This is attributed to the unique characteristics of the jurisdictional industry served by each code.

To show the function of these tables, we use ASME B31.3 tables as shown in Tables 3.1 and 3.2 as an example. The B31.3 table is somewhat more complicated than the others. It shows in-plane and out-plane categories of SIFs for each type of component. The purpose is to apply different stress intensification on different orientation of moment. These tables are used as follows. For each component, we first have to calculate the flexibility characteristic, h. From this flexibility characteristic, the flexibility factor and SIFs are calculated. It is clear from the table that for all the different components only the flexibility characteristics are calculated differently. The flexibility factor and SIFs are calculated more or less the same way for all components. This is the nice thing about the equivalent bend approach mentioned previously. The flexibility factor is included in structural analysis to obtain piping forces and moments. SIFs are then applied to piping moments to calculate pipe stresses.

ASME B31.1, on the other hand, provides only one stress intensification for each component. The value used is equivalent to the greater of the in-plane and out-plane values. Because of this one stress intensification approach, it is not necessary to distinguish in-plane moment or out-plane moment. The moments are considered the same regardless of their orientation. In fact, B31.1 stress intensification is also applicable to torsion moment, which is not applied with any stress intensification by B31.3 and some other codes. More on stress calculation is given in the next chapter, which deals with code stress requirements.

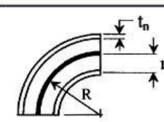
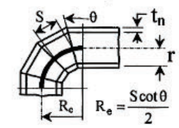
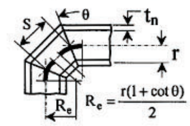
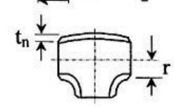
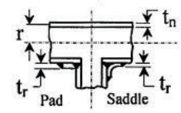
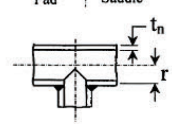
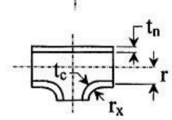
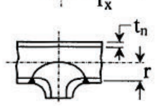
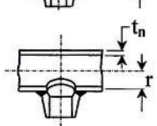
The stress intensification and flexibility factors at elbows and bends are sensitive to flanged ends and internal pressure. The effect of flanged end is discussed below and the effect of internal pressure will be discussed later in Section 3.7.

Effect of flanges on bend flexibility and SIFs. The flexibility factor and SIF at bends are mainly due to ovalization of the cross-section. Therefore, it is natural to expect these factors to be reduced by the stiffening effect of the flange connections. ASME code stipulates that when one or both ends of a bend are attached with flanges, the bend flexibility factor and SIF shall be multiplied by the factor C, where $C = h^{1/6}$ is for one end flanged and $C = h^{1/3}$ is for both ends flanged. For simplicity, it does not matter which end is flanged when only one end is flanged. The reduction factor is uniformly applied to the whole bend regardless of which end is flanged.

3.5 ALLOWABLE THERMAL EXPANSION STRESS RANGE

As discussed in the previous section, the failure mode of thermal expansion is fatigue failure requiring many cycles of repeated operations. In evaluating fatigue failure, the stress range encompassing the minimum and the maximum stresses in each cycle has to be considered. This is the reason why

TABLE 3.1
B31.3 FLEXIBILITY FACTORS AND STRESS INTENSIFICATION FACTORS [6]

Description	Flexibility factor, k	Stress intensification factor: Out-plane, i_o	Stress intensification factor: In-plane, i_i	Flexibility characteristic, h	Sketch
Welding elbow or pipe bend	$\frac{1.65}{h}$	$\frac{0.75}{h^{2/3}}$	$\frac{0.90}{h^{2/3}}$	$\frac{t_n R}{r^2}$	
Closely spaced miter bend $s < r\,(1 + \tan\theta)$	$\frac{1.52}{h^{5/6}}$	$\frac{0.90}{h^{2/3}}$	$\frac{0.90}{h^{2/3}}$	$\frac{\cot\theta}{2} * \frac{t_n s}{r^2}$	
Single miter bend or widely spaced miter bend $s \geq r\,(1 + \tan\theta)$	$\frac{1.52}{h^{5/6}}$	$\frac{0.90}{h^{2/3}}$	$\frac{0.90}{h^{2/3}}$	$\frac{1 + \cot\theta}{2} * \frac{t_n}{r}$	
Welding tee per ASME B16.9	1	$\frac{0.90}{h^{2/3}}$	$\frac{3}{4}\, i_o + 1/4$	$3.1\frac{t_n}{r}$	
Reinforced fabricated tee with pad or saddle	1	$\frac{0.90}{h^{2/3}}$	$\frac{3}{4}\, i_o + 1/4$	$\frac{\left(t_n + \frac{t_r}{2}\right)^{5/2}}{r\, t_n^{3/2}}$	
Un-reinforced fabricated tee	1	$\frac{0.90}{h^{2/3}}$	$\frac{3}{4}\, i_o + 1/4$	$\frac{t_n}{r}$	
Extruded welding tee	1	$\frac{0.90}{h^{2/3}}$	$\frac{3}{4}\, i_o + 1/4$	$\left(1 + \frac{r_x}{r}\right)\frac{t_n}{r}$	
Welded-in contour insert	1	$\frac{0.90}{h^{2/3}}$	$\frac{3}{4}\, i_o + 1/4$	$3.1\frac{t_n}{r}$	
Branch welded-on fitting (integrally reinforced)	1	$\frac{0.90}{h^{2/3}}$	$\frac{0.90}{h^{2/3}}$	$3.3\frac{t_n}{r}$	

stress range — rather than static stress — is considered in evaluating thermal expansion and other displacement stresses. As for the allowable stress, suggested values and their rationales have been thoroughly discussed by Rossheim and Markl [11, 13]. Their suggested values have been used by the codes [6] since 1955 with only minor intermittent modifications.

As discussed in Section 3.3, self-limiting stress dose not cause the abrupt gross structural deformation when it reaches the yield strength of the pipe. Therefore, if allowed by fatigue, the stress range

TABLE 3.2
FLEXIBILITY FACTOR AND STRESS INTENSIFICATION FACTOR FOR JOINTS [6]

Description	Flexibility factor, *k*	Stress intensification factor, *i*
Butt welded joint, or weld neck flange	1	1.0
Double welded slip-on flange	1	1.2
Fillet welded joint, or socket weld fitting	1	1.3–2.1
Lap joint flange (with B16.9 lap joint stub)	1	1.6
Threaded pipe joint, or threaded flange	1	2.3
Corrugated straight pipe, or corrugated or creased bend	5	2.5

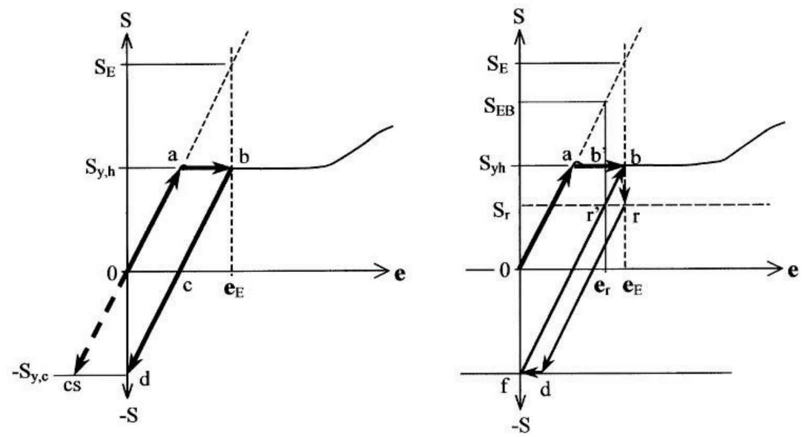
can exceed the yield strength. The starting point of investigation is an elastic equivalent stress range at twice of the yield strength. (The sum of the hot yield strength plus the cold yield strength, instead of twice of the yield strength, would have been a more accurate statement here.) This is also equivalent to a strain range, e_E, of twice the yield strain. The background for setting the allowable stress range can be explained by the help of three possible operating conditions (Fig. 3.7), described as follows:

(1) *No cold spring and no stress relaxation*. As shown in Fig. 3.7(a), this piping system heats up from the zero stress, zero strain point, 0, expanding gradually to reach the yield point at hot condition, *a*, and continues on to the final point, *b*. This final point corresponds to e_E with an elastic equivalent stress of the sum of the cold and hot yield strengths. At moderate operating temperature with no, or very little, stress relaxation expected, the piping stays at point *b* for the whole operating period.

When the system cools down, the pipe contracts, elastically reducing the stress from point *b* to point *c* where stress is zero, but the cooling process is only halfway complete. As the cool-down continues, the sign of the stress reverses. An initially tensile stress becomes compressive from this point on. The operation cycle ends at point *d* when the temperature returns to the pre-operation temperature. The total contraction strain is the same as the total expansion strain. The final pipe stress is equal to the cold yield strength, assuming the compressive stress-strain curve is the same as the tensile one. The next operating cycle starts from point *d*, and goes elastically to point *b* without producing any yielding. The subsequent operating cycles all follow the elastic line from *d-b* and back with *b-d*. Except for the initial yielding created by the first operating cycle, no yielding is produced in any subsequent operating cycle. This could signify that an expansion stress range below the sum of hot yield strength and cold yield strength probably would not produce a fatigue failure for an ideal material.

When the pipe cools down to ambient condition at point *d*, the system retains a reversed stress equivalent to the cold yield stress. This stressing at cold condition is equivalent to the effect of a cold sprung system. This situation is called *self-spring*. Self-spring causes flanges to suddenly spring apart as the last bolts are removed, on a line being disconnected after a period of service. A gap is created between the flanges when the line is disconnected. The gap is equivalent to the cold spring gap that will be discussed in Section 3.6, although the line was not initially cold sprung.

(2) *Fifty percent cold spring with no stress relaxation*. Because it is generally desired that no yielding in the piping is produced during the cold spring process, a 50% or less cold spring is done for an expected expansion stress range that is twice the yield strength. Figure 3.7(a) shows the case with 50% cold spring. The piping is first cut short to form a gap that is equal to 50% of the



(a) Stress beyond yield point

(b) Stress relaxation

FIG. 3.7
STRESS BEYOND YIELD POINT, COLD SPRING, AND STRESS RELAXATION

expected expansion, and then the ends of the gap are pulled together to pre-stress the system. With the expansion stress range of twice the yield strength as the benchmark, a 50% cold spring will produce a reverse stress equal to the cold yield strength at point cs. The system starts operating from this point, cs, elastically in a straight line all the way to the hot yield point, *a*, without creating any yielding on the pipe. The stress at this final point is one-half of the expansion stress range. The system cools down elastically from point *a*, again back to original point cs. The same cycle repeats through the life of operation without creating any yielding on the pipe.

By comparing line cs-*a* and line *d*-*b*, a cold sprung system operates essentially the same as a non-cold-sprung system, except without the initial yielding. Because the initial yielding of the non-cold-sprung system has very little effect on fatigue, cold spring has no effect on fatigue. Therefore, no credit on the expansion stress reduction is allowed for cold spring. The expansion stress range is the one that is counted.

(3) *With stress relaxation*. In high temperature environments, the stress relaxes with time when yielding or creep occurs in the pipe. Figure 3.7(b) shows a non-cold-sprung system with a stress relaxation at operating condition. As given in case (1), the operation reaches point *b*, at operating temperature. Owing to temperature effect, the stress gradually relaxes to point *r* while operating. When the pipe cools down, both stress and strain reduce elastically from point *r* to point *d* to reach the cold yield strength before the cool-down is completed. As the cool-down continues, it produces plastic strain *d*-*f* to reach the ambient temperature. In the subsequent operating cycle, the process follows *f*-*b*-*r*-*d*-*f*, again producing a *d*-*f* plastic strain. Because our goal is to avoid producing plastic strain in every cycle, the cold yield strength plus hot yield strength benchmark stress range is not suitable for situations with stress relaxation.

To prevent the generation of plastic strain in every operating cycle under stress relaxation condition, the benchmark stress range has to be reduced to cold yield strength plus the stabilized relaxation residual stress, S_r. This benchmark stress range is shown as S_{EB} in Figure 3.7(b). With this modified benchmark stress range, the pipe will operate along *f*-*r*′ pass indefinitely. However, the problem is that the stabilized relaxation residual stress is not readily available for most materials.

From the preceding deductions, the benchmark allowable expansion stress range can be set as the sum of the cold yield strength and hot yield strength at below creep range. At creep range, it will be the sum of the cold yield strength and the creep strength at expected plant life. Markl [11] suggested that it appears to be conservative to make it the sum of the cold yield strength and 160% of the stress producing 0.01% creep in 1000 hours at the operating temperature. The benchmark stress range can be written as

$$S_{EB} = (S_{yc} + S_{yhx}) \tag{3.13a}$$

where

S_{yc} = yield strength at cold condition
S_{yhx} = is the lesser of the yield strength at hot condition and 160% of the stress producing 0.01% creep in 1000 hours at the operating temperature

The stress producing 0.01% creep in 1000 hours is one of the criteria of setting the hot allowable stress. In terms of ASME B31.1 allowable stresses, which was originally set at no greater than 5/8 of the yield strength at corresponding temperature, the above equation can be written as

$$S_{EB} = 1.6(S_c + S_h) \tag{3.13b}$$

where S_c is cold allowable stress and S_h is hot allowable stress. The benchmark stress range, S_{EB}, is considered the maximum stress range to which a system could be subjected without producing plastic flow at either cold or hot limit. In ASME B31.3 code and the recent B31.1 code, because of their use of higher allowable stress based on 2/3 of the yield strength, the benchmark stress range has become

$$S_{EB} = 1.5(S_c + S_h) \tag{3.13c}$$

The above derivation of the non-yielding benchmark stress range requires further justification before being used. The first thing that needs to be clarified is the fact that the code stress is only one-half of the theoretical stress when an SIF is involved, as discussed in Section 3.4. Therefore, theoretically the constants in Eqs. (3.13b) and (3.13c) have to be halved to avoid any yielding. However, because only the local peak stress and shell bending stress are halved, the difference between the code stress and the theoretical stress affects only the final fatigue evaluation. Equations (3.13b) and (3.13c) ensures that the main load resisting membrane stress would not exceed the yield. This is important in avoiding gross structural deformation, and in ensuring the validity of elastic analysis commonly used.

The other thing that needs to be worked out is the actual application of the benchmark stress range. This mainly involves the setting of the safety factor. If the ASME B31 code had used the theoretical SIF, the benchmark stress range would have represented the stress range limit for unlimited operating cycles. In actual applications involving limited number of operating cycles, Markl suggested a total allowable stress range of

$$S_A + S_{PW} = 1.25(S_c + S_h) \tag{3.14}$$

where S_A is the basic allowable stress range for calculated thermal expansion stress and S_{PW} is the sustained stress due to pressure and weight. By comparing with Eqs. (3.13b) and (3.13c), this represents 78% and 83% of the benchmark stress range. This may not be very significant as we are comparing the calculated stress with a theoretical stress. However, the important thing is whether this allowable stress provides enough safety factor for systems operating at limited number of cycles. Because the longitudinal stress due to pressure and weight is generally allowed to reach hot allowable stress, that is, $S_{PW} = S_h$, the allowable stress range for thermal expansion only, S_A, becomes

$$S_A = f(1.25S_c + 0.25S_h) \tag{3.15}$$

where f is a stress-range reduction factor varying from $f = 1.0$ for $N < 7000$ cycles, to $f = 0.5$ for $N > 250{,}000$ cycles. However, the newer fatigue data has shown that the f value may be considerably smaller than 0.5 when $N > 250{,}000$ cycles. The exact value used by the code will be discussed in another chapter dealing with code design requirements. See also Chapter 13 on vibration analysis for high cycle fatigue. The stress-range reduction factor, f, has been recently renamed stress range factor because it reduces the allowable stress, and not the stress range.

Based on the experimental data, the allowable expansion stress range given in Eq. (3.15) represents an average safety factor of 2 in terms of stress, and a safety factor of 30 in terms of cyclic life. However, because of the spread in individual test data, the potential minimum safety factor can be as low as 1.25 in terms of stress and 3 in terms of life. This emphasizes the need for making a conservative estimate of the stress and the number of operating cycles. It should be noted, however, that these safety factors were based on the original (1955) allowable stress, which was taken as $^5/_8$ of the yield strength instead of the current $^2/_3$ of the yield strength.

The stress-range reduction factor cuts off at 7000 cycles with no increase in allowable stress-range permitted when the operating cycles are less than 7000. With f=1.0 at 7000 cycles, the allowable stress has already reached the benchmark stress limit. Any stress beyond that might produce gross yielding in the system, thus invalidating the elastic analysis. Therefore, the actual safety factor increases as the number of cycles reduces. The 7000 operating cycles represents 1 cycle/day for 20 years. This number is more than most piping systems experience nowadays when batch-operating processes requiring daily turnaround are rare.

Some engineers may be puzzled that the allowable stress range as given in Eq. (3.15) can reach as high as the ultimate strength of the pipe material. We want to re-emphasize here that the stress calculated and referred to is the elastic equivalent stress as discussed in Section 3.3. The actual stress is always less than the yield strength of the material. It is also be emphasized that corrosion in the pipe can substantially reduce fatigue life at an unpredictable rate. Therefore, proper control of corrosion with some adjustment of the allowable stress may be required. Furthermore, when a piping system is connected to delicate equipment, such as a pump, turbine, or compressor, the allowable stress is governed by the allowable reaction forces and moments of the equipment. The allowable equipment reaction corresponds to the static stress, which may be reduced by cold spring. The allowable stress for the piping connected to a piece of delicate equipment can be as small as only a fraction of the allowable stress range given in Eq. (3.15).

3.6 COLD SPRING

Cold spring, pre-spring, and cold pull all refer to the process that pre-stresses the piping at installation or cold condition in order to reduce the force and stress under the operating or hot condition. Cold spring is often applied to a piping system to: (1) reduce the hot stress to mitigate the creep damage [14]; (2) reduce the hot reaction load on connecting equipment; and (3) control the movement space. However, at the creep range the stress will be eventually relaxed to the relaxation limit even if the pipe is not cold sprung. The general belief is that the additional creep damage caused by the initial thermal expansion stress is insignificant [11] if the total expansion stress range is controlled to within the code allowable limit. The real advantage of cold spring has become the reduction of the hot reaction on the connecting equipment. For instance, a 50% cold spring would split the reaction force into halves. One-half of it is realized at cold condition as soon as the pipe is cold sprung. The more critical hot reaction is reduced to only one-half of the total reaction that otherwise would have been imposed by a non-cold-sprung system.

Although cold spring offers unquestionable benefits, its adoption varies among industries. In power plant piping, cold spring is done normally for most of the major piping systems such as main steam, hot reheat, and cold reheat lines. On the other hand, cold spring is seldom used in petrochemical process piping. The petrochemical industry shies away from cold spring mainly due to the belief that the intended benefits of cold spring may not be realized due to inadequate specifications and poor supervision of field erection procedures. This has become the culture of the two industries partly due to the much larger amount of piping involved in a petrochemical plant than in a power plant. Without proper specifications and installation procedures, no cold spring should be attempted. In some cases, even the self-sprung gap is closed by cutting-and-pasting the piping rather than cold springing it back for connection. Nevertheless, even in the petrochemical industry, a one-dimensional cut-short on yard piping to control the corner movement is occasionally performed. This cut-short on yard piping requires no special procedure due to the small force involved in a very flexible system.

The cold spring process involves laying out the piping somewhat shorter than the installing space to create a gap at the final weld or final bolting location, as the system is erected without straining. The system is then pulled or pushed according to a predetermined procedure to close the gap and to finish the final joint.

3.6.1 Cold Spring Gap

The size of the cold spring gap depends on the cold spring factor desired. The size of the gap for a 100% cold sprung system is the total amount of system expansion modified with the differential anchor movement. For example, take the simple system shown in Fig. 3.8, where a three-dimensional gap is generally required to achieve a uniform cold spring. Gap size can be calculated form the overall dimensions, L_x, L_y, and L_z, plus the differential effect of the anchor displacements. The procedure can be quite confusing in a general thee-dimensional system. Therefore, it is recommended to use a systematic procedure that does not require visual interpretation. Before calculating the gap, we need to define the gap orientation first. This gap orientation is required for entering the cold spring gap data into a computer program. Generally, the gap is considered as running in the same direction as the sequence of node points. In this sample piping system, we are describing the system from point A to point B; therefore, the gap is running from point G_A to point G_B. A minus gap means running from G_A to G_B in the negative coordinate direction. For the x direction, the gap is calculated by

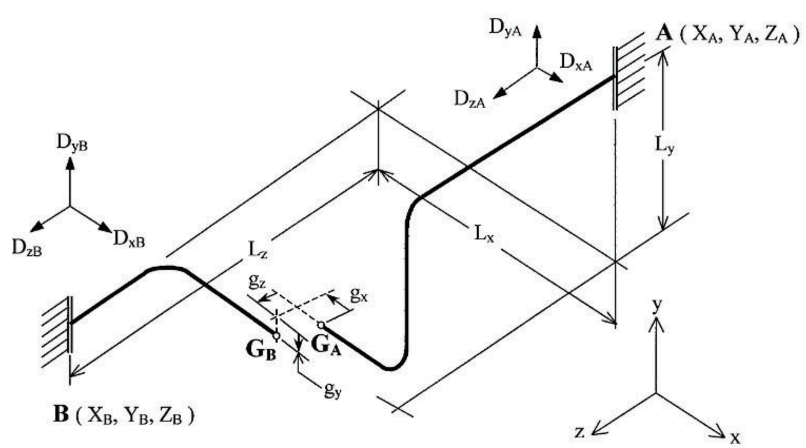


FIG. 3.8
COLD SPRING GAP

$$g_x = C\{\alpha(T_2 - T_1)(X_B - X_A) - (D_{xB} - D_{xA})\}$$

where C is the cold spring factor ranging from 0.0 for no cold spring to 1.0 for 100% cold spring; X_A, X_B are x coordinates for anchor A and anchor B, respectively; and D_{xA} and D_{xB} are x direction anchor displacements for anchor A and anchor B, respectively. The gaps in y and z directions are calculated with similar procedures.

The case assumes that the piping is operating at a temperature higher than the construction temperature. A 100% cold sprung system, if installed properly, will have the expansion stress reduced to zero when the system reaches the operating temperature. It will be free of any thermal expansion stress under the hot operating condition. This assumes that the stress range is less than the yield strength. For cases where stress range is greater than yield strength, the reduction of reaction and stress at hot condition is not that straightforward.

3.6.2 Location of Cold Spring Gap

The gap can be located anywhere in the system with the same gap size. However, it is important to locate the gap at the place where it is easy to work and achieve the intended result. Unless other factors have demonstrated otherwise, the gap is normally placed at the location near one of the terminal anchors. This allows the entire line, except a small part at the anchor side, of flexibility to be available for closing the gap. No pulling is generally required on the short side of the line. The free end can be simply clamped down into position. Locating the gap close to the terminal end also ensures a minimum rotation of the pipe. Besides the cited benefits of locating the gap at points near one of the terminal ends, the accessibility and the amount of force required are also consideration factors. From the standpoint of the forces and moments required, the most desirable location is the one that produces the minimum internal pipe forces and moments, especially moments under operating temperature without cold spring. Because the cold spring process essentially has to create the reverse mirror image of the operating condition, a gap at a high internal force and moment location requires a large pulling effort to create the proper cold spring. The most difficult location is the one that requires a large moment, especially the torsion moment twisting around the pipe axis. An analysis of the operating condition needs to be performed so the proper location can be selected. Another factor that needs to be considered is whether there is enough working space for pulling and welding.

3.6.3 Cold Spring Procedure

The success of cold spring depends on the specification and execution of the procedure. Preparation of cold spring starts with the assignment of the cold spring gap. Next, a set of drawings is prepared showing the fabrication dimensions versus the routing dimensions of the piping. The hanger and support attachment locations are laid out in reference to the final cold sprung locations. During the erection of the cut-short piping, a spacer equivalent to the cold spring gap is generally clamped to the ends of the gap to ensure the proper gap size. A marking line in the axial direction is drawn across the gap on both ends of the pipe. This line is used to ensure the proper alignment against the twisting between the ends. Based on the magnitude of the forces and moments expected at the gap location, some strategic locations are selected for applying the pulling forces. In general, more than one pulling locations are needed for each side of the piping.

Before starting the cold spring, an analysis with the gap applied and piping at ambient temperature has to be performed. This analysis includes the piping on both sides of the gap. Displacements from the analysis are the movements expected from the cold spring. Displacements near each end of the gap show the amount of cold spring required from each side of the piping. The expected cold spring movements are then noted on the drawing. These movements are subsequently marked in the field as guides for the pulling. An analysis for each side of the piping, with the cut-off at the gap, can also be performed to estimate the pulling forces required. This analysis is accomplished by applying a unit force on each pulling location in the selected direction to see the amount of the gap movement due to each force. A proper combination of forces can then be determined by matching the total movements against the cold spring required on each side of the piping. As a rule of thumb, a three-dimensional cold spring gap generally requires three pulling forces located at strategic locations. The directions of these pulling forces are more or less perpendicular to each other. Additional effort may be required to prevent twisting between two ends. During the pulling, in addition to the permanent hangers and locked spring hangers, some temporary supports may also be required. As the pipe moves up and down, the hanger rods need to be adjusted accordingly to match the expected cold spring movements. The final welding is performed with both sides of the piping clamped together at the gap.

All spring hangers and rod hangers are adjusted after the cold spring is completed. They are adjusted in the same way as would be done had the piping not been cold sprung. The spring hangers are locked at their regular cold positions until the piping is hydro-tested and ready to operate.

3.6.4 Multi-Branched System

Cold spring for a multi-branched system is considerably more complicated than that for a two-anchor system. In general, more than one gap is required to achieve a uniform cold spring [15] of a multi-branched system. This would require a close integration of design, analysis, fabrication, and erection activities. The procedure can overwhelm even a well-organized company. Therefore, one alternative is to cold spring just the main portion of the piping instead of uniformly cold springing the entire system. From the basic beam formula, we know that the force generated by a displacement is inversely proportional to the cube of the beam length that is perpendicular to the direction of the displacement. The force generated by the expansion of L_x is inversely proportional to roughly the cube of the resultant length of L_y and L_z. Similarly, the forces generated by the expansion of L_y and L_z are inversely proportional to the cube of the resultant lengths of (L_x, L_z) and (L_x, L_y), respectively. This means that the longest leg in a system dominates the response of the entire system. The long leg produces the large expansion, which has to be absorbed by the short perpendicular legs in the system. Because these short perpendicular legs provide very limited flexibility, very high forces and stresses are generated in the system mainly due to just one long pipe leg. Thus, a cold spring only in the direction of the longest leg serves approximately the same purpose as a uniform cold spring in all three directions.

Figure 3.9 shows a simple three-branch system that needs to be cold sprung to reduce the anchor reaction. If a uniform cold spring were required, three cold spring gaps would have to be provided and complicated procedures would have to be developed and executed, which would be a real challenge to accomplish. Instead, because the major leg length is in the x direction, we can simply cold spring the x direction with the gap as shown. This will achieve the main purpose of the cold spring with a much simpler procedure.

3.6.5 Analysis of Cold Sprung Piping System

The piping code has specifically stipulated that the cold spring cannot be credited for reducing the expansion stress, because the main failure mode of the displacement stress is the fatigue failure, which is dictated by the stress range. A cold spring will shift some stress from the hot operating condition to the cold condition, but the stress range remains unchanged. The code further stipulates that although cold spring can reduce the hot reaction to the connecting anchor or equipment, only $^2/_3$ of the theoretical reduction can be taken. This is because cold spring is a very complicated procedure. A perfect cold spring is seldom achieved.

For a uniformly cold sprung system, an analysis under operating conditions is all that is required to obtain all the reaction information. However, because a uniform cold spring is seldom performed

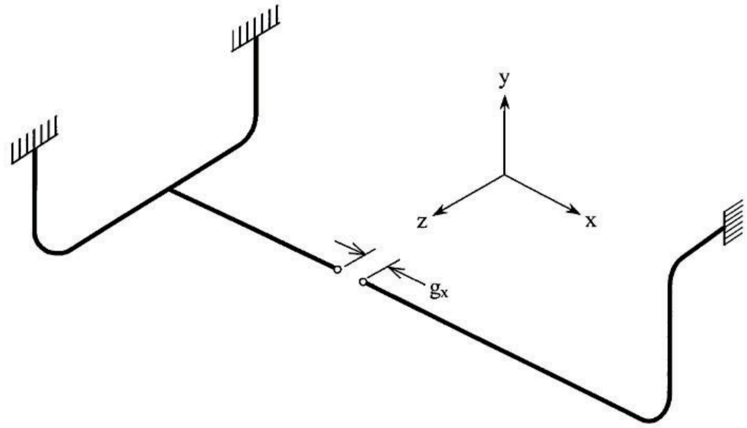


FIG. 3.9
LOCAL COLD SPRING

in actual installations, more analyses are required. In general, the following three analyses regarding thermal expansion are required:

(1) Under operating temperature, but no cold spring gap. This is used to check the thermal expansion stress range to compare with the code allowable stress range.
(2) Under operating temperature with $^2/_3$ of the cold spring gap. This is used to find the credible anchor reaction at operating condition. The $^2/_3$ factor is the fraction of the cold spring effect allowed by the code.
(3) Under ambient temperature with full cold spring gap. This is used to find the anchor reaction at the cold condition after the cold spring. This analysis is also used to provide displacement guides for executing the cold spring process.

These three analyses can be performed one by one, or all together, in an integrated computer analysis. Although the piping code does not stipulate the uncertain pulling effect on the cold reaction, it should be noted that due to less-than-perfect pulling, a much higher cold reaction than the one given by analysis (3) above may be obtained.

3.7 PRESSURE EFFECTS ON PIPING FLEXIBILITY

Because of its nature as a sustained load, internal pressure is often ignored in the evaluation of piping flexibility. This tendency has been rectified recently, but is still not widely appreciated. For instance, pressure is also a cyclic load that needs to be included in the fatigue evaluation. This cyclic pressure, which is implied in Eq. (3.14), is seldom appreciated. Furthermore, pressure also has very significant effects on piping flexibility itself. Flexibility effects occur in two areas: (1) pressure elongation of the piping element and (2) pressure effect on the bend flexibility factor and bend SIFs.

3.7.1 Pressure Elongation

In a pressurized pipe, the entire inside surface of the pipe sustains a uniform pressure. This pressure loading develops a tri-axial stress in the pipe wall. The stress component normal to the pipe wall is generally small and will be ignored in this discussion. The pipe wall is subjected to stresses S_{hp} and

S_{lp} in circumferential and longitudinal directions, respectively, as shown in Fig. 3.10. Stress in the circumferential direction is generally referred to as hoop stress. For practical purposes, S_{lp} is equal to one-half of S_{hp}, as shown in Eqs. (2.13) and (2.14). The hoop stress and the longitudinal stress generate strains in all major directions. The relation between stress and strain is given by Eq. (2.8). By substituting $S_{lp} = 1/2S_{hp}$, the strains in hoop and longitudinal directions become

$$e_H = \frac{S_{hp}}{E} - \nu\frac{S_{lp}}{E} = \frac{S_{hp}}{E}(1 - 0.5\nu) \tag{3.16}$$

$$e_L = \frac{S_{lp}}{E} - \nu\frac{S_{hp}}{E} = \frac{S_{hp}}{E}(0.5 - \nu) \tag{3.17}$$

Elongation in the hoop direction increases the pipe diameter. This diametrical change progresses freely without creating any additional resistance to the piping system. Elongation in the longitudinal direction, on the other hand, behaves like thermal expansion. This elongation, given by Eq. (3.17), has a squeezing effect on the piping system just like thermal expansion.

The significance of the pressure elongation can be better visualized by converting it to an equivalent temperature rise [16]. Figure 3.10 shows the relationship between pressure elongation in terms of pressure hoop stress and its equivalent temperature rise, for low carbon steel pipe. For plant piping, hoop stress is normally maintained at less than 15,000 psi (103.4 MPa). In this case, pressure elongation is equivalent to a temperature increase of 17.5°F (9.72°C). In view of the high temperature range normally experienced in plant piping, the effect of pressure elongation in plant piping is insignificant. On the other hand, hoop stress in a cross-country transportation pipeline can reach 30,000 psi (206.8 MPa) or higher for high-strength pipe. The equivalent temperature of this hoop stress range can easily exceed 35°F (19.44°C), which represents a high percentage of the design temperature rise. Therefore, pressure elongation is an important factor to be considered in transportation pipeline design.

3.7.2 Potential Twisting at Bends

The longitudinal elongation discussed in the preceding section is also applicable to piping bends, but the curvature at the bends might also produce different type of deformations. One of the impressions that engineers would intuitively form is that the bend will tend to open when pressurized. Equations have been developed for this opening rotation and are implemented in some computer programs. This opening effect is often misquoted as the Bourdon tube effect. It has artificially generated, on

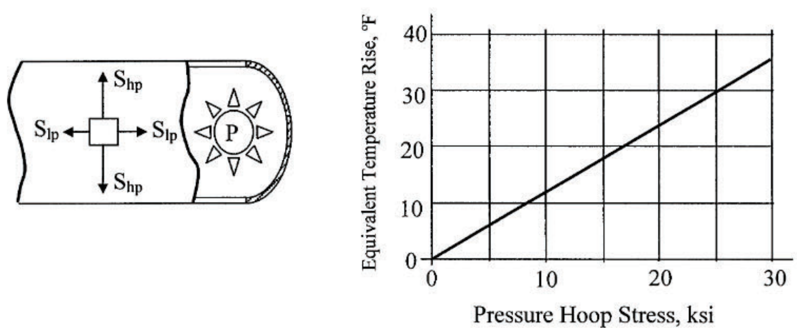


FIG. 3.10
PRESSURE ELONGATION

paper, much bigger twisting effects on the piping than was experienced in the field. Actually, a Bourdon tube is an arc shape tube with an oval cross-section. This oval cross-section is the key for the opening rotation. The general belief in the piping community on piping bends is that if the cross-section out-of-roundness is insignificant, the rotation of the bend is not expected [17].

Detailed analyses and tests on special miter bends [18] have shown that in the elastic stress range, the bends actually close rather than open after pressurization. It starts to open only after the pressure stress exceeds the yield strength. It appears that until more rigorous theories and tests prove otherwise, the opening of the pipe bend by the pressure shall be ignored.

3.7.3 Pressure Elongation Is Self-Limiting Load

Pressure elongation is often mistreated as a sustained load because of its association with pressure, which is a sustained load. Just like thermal expansion, pressure elongation generates a displacement that is a self-limiting load. Its effect on the piping system is determined by the potential axial displacement of each leg of the piping. Once the displacement reaches the potential elongation amount, it stops regardless of whether or not the yielding occurs in the piping. This pressure elongation is generally included in the flexibility analysis the same way as thermal expansion is. In general, pressure elongation is added to thermal expansion to become the total displacement load in the analysis.

3.7.4. Pressure Effect on Bend Flexibility and SIFs

The flexibility factor and SIF at a bend are mainly caused by the ovalization of the bend cross-section. Internal pressure tends to reduce ovalization, thus reducing flexibility and stress intensification. The pressure effect on bend flexibility and SIFs has been well investigated. One of the most recognized treatments is the use of modification factors established by Rodabaugh and George [17]. These factors have been adopted by the ASME piping codes, and are summarized bellow.

In large-diameter, thin-wall elbows and bends, pressure can significantly affect the magnitudes of the flexibility factor, k, and the SIF, i. Under pressurized conditions, the value of k calculated from Eq. (3.5) should be adjusted by dividing it with

$$\left[1 + 6\left(\frac{P}{E}\right)\left(\frac{r_m}{t_n}\right)^{7/3}\left(\frac{R}{r_m}\right)^{1/3}\right] \tag{3.18}$$

And the SIFs calculated by Eqs. (3.11) and (3.12) may be reduced by dividing them with

$$\left[1 + 3.25\left(\frac{P}{E}\right)\left(\frac{r_m}{t_n}\right)^{5/2}\left(\frac{R}{r_m}\right)^{2/3}\right] \tag{3.19}$$

where

r_m = mean radius of the pipe
t_n = nominal thickness of the pipe
R = radius of the bend curvature

The use of Eqs. (3.18) and (3.19) varies from case to case. Some specifications require the application of the flexibility reduction factor given by (3.18), but not the stress intensification reduction factor given by (3.19). The rationale is that piping tends to maintain its temperature longer than it can maintain its pressure during shutdown. In other words, the stress intensification reduction factor may disappear while the piping is still under full expansion state.

3.8 GENERAL PROCEDURE OF PIPING FLEXIBILITY ANALYSIS

In previous sections, we have discussed the fundamentals of piping flexibility. In this section, we will focus on the general procedure for performing the piping flexibility analysis. A computer program is generally used in the analysis. The simple system shown in Fig. 3.11 will be used to explain the general procedure.

Although most computer program will check the adequacy of the pipe wall thickness against design pressure, wall thickness is actually determined before the analysis is performed. Before the analysis, isometric drawings, together with all the data, (e.g., pipe material, diameter, thickness, design pressure, design temperature, and upset temperature), have to be collected. This data is generally contained in the line list and material specification. The analysis covers the following steps.

3.8.1 Operating Modes

The system may have several different operating modes. Although it is possible to select one mode that represents the most critical situation, normally more than one operating mode has to be considered. In this sample system, there are three possible operating modes: (1) both heat exchanger loops are operating; (2) only HX-1 heat exchanger is operating; and (3) only HX-2 heat exchanger is operating. The analyst decides which of these three modes needs to be analyzed. At first sight, it may appear

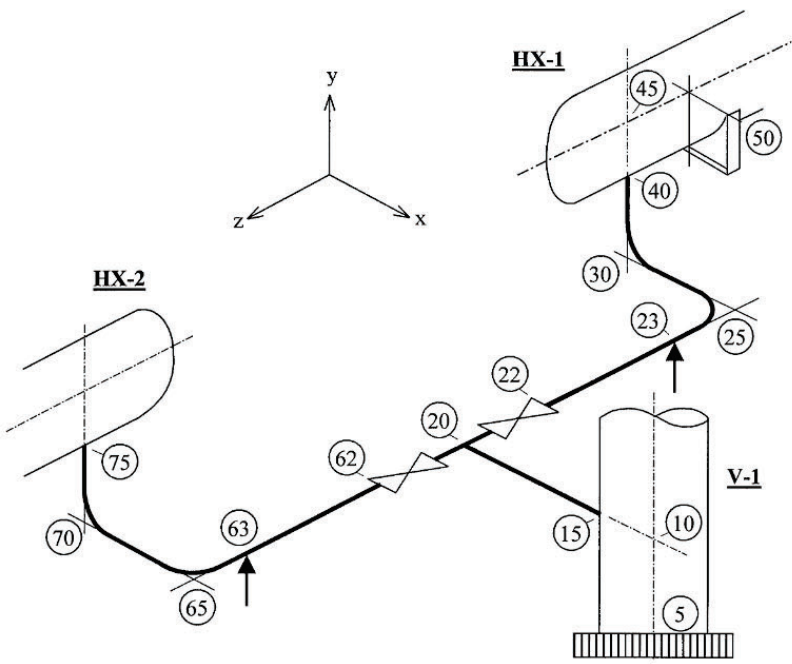


FIG. 3.11
SAMPLE ANALYSIS PROCEDURE

that only the first mode needs to be analyzed. However, a detailed investigation will show that all three modes are required, and they have to be analyzed in one integrated computer run.

It is expected that the first operating mode with both loops operating will result in the highest stress and reaction load at heat exchanger connections 40 and 75. This first operating mode, however, essentially generates no stress and load at vessel connection 15. In the second operating mode with only HX-1 operating, the tee point 20 will be pushed toward HX-2 producing significant stress and load at point 15. Symmetrically, the third mode, with HX-2 only operating, will produce the same amount of stress and load as mode 2 at point 15, but in a reverse direction. Because the stress range at point 15 is the difference of the stresses generated from operating modes 2 and 3, analyses on both modes are required. Differential moments from both operating modes are taken to calculate the stress range required for expansion stress evaluation. The stress range cannot be calculated by subtracting the calculated stresses. This is due to the normal practice of calculating the effective stresses and stress intensities, which have lost the sign of the stress.

3.8.2 Anchor Movements

Points 15, 40, and 75 are terminal points that can be considered anchors or vessel connections. Either way, there are movements at these points attributed mainly to the expansion of the vessels. These movements are calculated by multiplying the corresponding active length with the expansion rate per unit length. Take point 15, for instance, where movement is produced by the expansion of the vessel shell from vessel anchor point 5 to point 15. In the *y* (vertical) direction, it has a movement equivalent to the expansion from point 5 to point 10. In the *x* direction, it has a movement equivalent to the expansion of 10-15. These movements are called free movements. They are the same as the final displacements at point 15 when an anchor is used. However, they can be slightly different from the final displacements at point 15 if a vessel connection is used. Vessel connections are discussed more fully in a separate chapter. The *x* movement in this case is negative, because it is moving in the negative *x* direction.

Anchor movements at points 40 and 75 are determined the same way as for point 15. However, for heat exchangers and horizontal vessels that are supported with two saddles, the support arrangement plays an important role in piping flexibility. The thermal expansion stress in piping differs considerably, depending on the selection of the fixed support point. To ensure positive control of the movement and to offer positive resistance to horizontal loads such as wind and earthquake, one of the saddle supports is generally fixed and the other is allowed to slide axially with slotted bolt holes. The vessel will then expand axially from the fixed saddle support. In this sample system, it is advantageous to fix the support at point 50, so that it is closer to connection point 40 to reduce the *z* direction movement. The *z* direction movement is in the positive direction, and is moving against the piping expansion. For the *y* movement, the actual zero-displacement location has to be estimated. In this case, it can be assumed to be at the bottom of the vessel shell. Because this elevation is roughly the same as the elevation of point 40, the *y* movement can be considered as zero. In this example, points 5, 10, 45, and 50 are not required for the analysis. They are used just for calculating the anchor displacements.

3.8.3 Assignments of Operating Values

The system has three major operating modes to be analyzed. This requires a proper assignment of pipe data groups and operating values. It is not difficult to realize that three groups of pipe and operating data are required. The first one covers the main line from point 15 to 20 continues on to point 22 and 62, the second one covers the first branch from point 22 to 40, and the third one covers the rest of the piping. The three modes of operation are defined by the assignment of the temperatures for each of the modes, on each set of pipe data. Each anchor or vessel connection point should also be assigned with three sets of anchor movements, each corresponding to one of the operating modes. For instance, if a given branch is not operating, then the corresponding movements can either be zero or something corresponding to the idle temperature of the heat exchanger shell.

The multi-operating mode analysis is required for the calculation of stress range when a stress reversal is likely to occur through the change of operating modes. It is preferable to analyze all modes in one computer run, so the signs of the moments between the modes can be automatically checked.

3.8.4 Handling of Piping Components

As discussed in Chapter 2, a piping system is generally treated as a series of straight pipe elements and curved pipe elements in the analysis. Therefore, all components in the system have to be simulated with these two types of elements. Furthermore, analyses are based on the centerline of the pipe, with all the lengths defined up to the centerline intersections. The manner in which the components are treated will be discussed in the text that follows, starting sequentially from point 15.

Point 15 is considered an anchor or a vessel connection. If it is assumed as an anchor, then the point is considered perfectly rigid, unless the stiffness or spring rate in any of the six degrees of freedom, three in translation and three in rotation, is specified. If it is treated as a vessel connection, then the magnitudes of the stiffness in all six directions are calculated based on the vessel data. In this case, the vessel diameter, thickness, and reinforcement dimensions have to be specified. The use of a vessel connection represents a more accurate analysis, and generally results in a smaller thermal expansion stress. More detailed discussions on vessel connections are given in a separate chapter. Regardless of the type of end simulation used, the same anchor movements have to be specified.

Point 20 is a forged welding tee, or a fabricated branch connection, as the case may be. The connecting pipe is considered as extending to the centerline intersection point, without considering the existence of the branch connection. The stresses are calculated at this centerline location using the data of the connecting pipe. The only distinction between the branch connection and the regular pipe is that in a branch connection a keyword or flag has to be placed to inform the computer to include the proper SIF. A more sophisticated analysis will consider the branch connection the same as a vessel connection for large thin pipes when the run and branch diameter ratio is greater than 3.

Point 22 is the end of a valve. The valves, flanges, and other fittings are considered regular straight pipe with an additional stiffness added. It is more or less a standard industry practice to consider the valves to be three times as stiff as the connecting pipe of the same length. The valve is therefore treated as a pipe with three times the stiffness. To increase the stiffness, EI, either wall thickness or modulus of elasticity can be increased. The increase in thickness will proportionally increase the section modulus of the pipe, thus artificially reducing the stress calculated. An increase in modulus of elasticity does not have this shortcoming and is the preferred approach.

Point 23 is a support point. A rigid support does not allow the pipe to move in the supporting direction, and thus may significantly reduce the system flexibility. Normally, it is preferred to support the piping system with rigid supports as much as possible. This is mainly due to economy as well as reliability of the system. The rigid support is inexpensive, easy to install, and offers positive resistance to occasional loads such as wind or earthquake. However, there are locations where rigid supports are not suitable due to potentially large loads and stresses that might be generated. A large calculated force at the support normally communicates to the engineer that the location is not suitable for a rigid support. In such cases, a spring support may be required. Occasionally, it may also mean that support at this location is not required at all. A more detailed discussion of supports is given in a separate chapter.

Point 25 is a bend. As discussed earlier in this chapter, the bend has added flexibility and SIFs as compared to a theoretical circular beam with a non-ovalizing circular cross-section. These factors are all automatically included in a computer program if the geometrical dimensions of the band are specified properly. Most computer programs require the dimensions of the tangent intersection point. Therefore, the data point is assigned at the tangent intersection point. From the geometrical information of the tangent intersection point and the bend radius specified, the program locates the beginning and ending points of the bend. These end points, which may be called 25a and 25b in some cases, are the points used in the analysis.

Point 40 is another terminal point that can either be assumed as an anchor or treated as a vessel connection. For heat exchangers, due to its smaller shell diameter, the connections are most often considered anchors. The discussion for point 15 is generally also applicable to points 40 and 75.

3.8.5 The Analysis

The information gathered (discussed above) has to be organized, in a format required by the computer program of choice, to become the input data. The first thing that needs to be done is to assign a global coordinate system that will allow all pipe elements and loads to refer to a common coordinate system. Most programs require that the y coordinate should be in the vertical direction pointing upward. This convention follows the tradition of piping practices. With the y-axis in the vertical direction pointing up, it has a direct relation with the elevation scale normally used in the plant piping drawings. It also simplifies the weight load calculation, as the weight direction is automatically tied to the negative y direction. With a modern desktop or laptop computer, an analysis should not take more than a few minutes to complete once the data is prepared. In general, it requires more than one analysis to reach a satisfactory design of the system.

3.9 PROBLEMS WITH EXCESSIVE FLEXIBILITY

Because everyone is aware of the importance of piping flexibility, engineers have a tendency to provide more flexibility than required in a piping system. Indeed, it is widely believed that the more flexibility that is provided, the more conservative is the design. This is actually a very serious misconception. Today, it is pretty accurate to say that more piping problems have been created by excessive flexibility than by insufficient flexibility.

One obvious consequence of excessive piping flexibility is the added cost due to additional piping provided and the plant space required to accommodate it. In addition to the obvious one, there are many less obvious, but serious, consequences. For example, a flexible system is weaker in resisting wind, earthquake, and other occasional loads. It is also prone to vibration. Although some restraints and snubbers can be used to increase resistance to occasional loads, the added cost can be very substantial. The nature of piping flexibility can be explained with a railroad-construction analogy. There are four situations encountered in piping flexibility that resemble the four types of railroad construction methods as shown in Fig. 3.12.

Figure 3.12(a) shows the case with the rails laid end to end without any clearance. When the weather gets hot and the rail wants to expand but cannot expand, it generates a large amount of force between the ends. This can eventually buckle the rails making it inoperable. This is similar to a situation where the pipe is lying straight between two anchors. This illustrates the case of no flexibility, which can cause problems in piping and connecting equipment.

Figure 3.12(b) shows the rails laid with a small gap provided between two joining ends. This gap allows the rails to expand into it without generating any load. With this small, but sufficient gap provided, the buckling due to thermal expansion is eliminated. However, with a gap, no matter how small, a slight bouncing occurs every time the wheel passes over the gap. The rail is safe, but the ride is somewhat bumpy. This is a compromised arrangement nobody should complain about. The same thing happens to a piping system with just enough loops and offsets for the required flexibility. The flexibility in the piping eliminates the potential damage due to thermal expansion. Meanwhile, the piping system also becomes a little shaky, but it is tolerable. This is the design we need, and can live with.

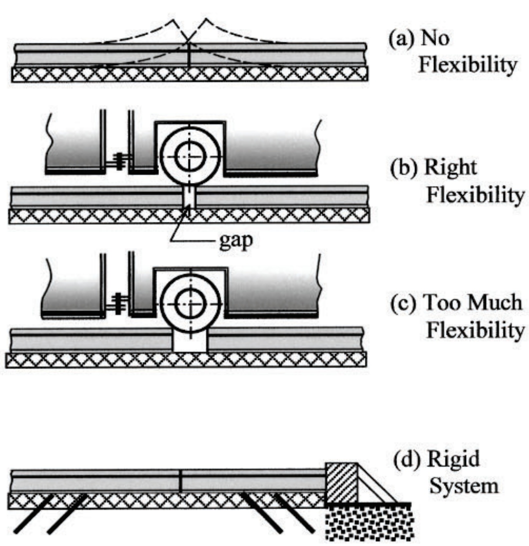


FIG. 3.12
RAILROAD ANALOGY TO PIPING FLEXIBILITY

Figure 3.12(c) shows the consequence of too much flexibility. Realizing that a proper gap between the ends of adjacent rails is necessary for the safety of the railroad, some engineers might get the idea of substantially increasing the gap size to enhance safety. When the gap becomes unnecessarily large, the potential damage due to thermal expansion is eliminated but the wheels of the railcar may not pass the gap safely. This creates a hazardous situation to the operation. In the design of piping systems, many design engineers believe that it is conservative to provide more flexibility in the piping. This indeed has become a guideline in some design specifications. However, just like the railroad with large gaps, this so-called conservative approach can also create operational hazardous in the piping.

Figure 3.12(d) shows a fully restrained continuously welded railroad construction. In a fully restrained situation, the rail generates about 200 psi/°F (2.48 MPa/°C) of axial stress due to temperature change. The stress is compressive if the operating temperature is higher than the construction temperature. With a maximum expected axial stress below about $^2/_3$ of the yield strength range, the railroad can be constructed continuously without gaps. This continuous construction allows for the smooth operation of the railcar when operating at high speeds. However, the thermal expansion problem is still there no matter what type of construction is adopted. In this type of construction, the thermal expansion problem is treated differently. It requires that heavy anchors be placed to prevent the ends of the rail from moving. Sufficient lateral restraints are also placed along the line to prevent the rail from lateral buckling. With all associated problems taken care of, the rigid railroad system offers a very smooth, safe operation for the railcars even at very high speeds.

There are similar situations in the construction of some piping systems. In cross-country pipelines, the fully restrained construction is generally adopted to reduce the cost and increase the reliability. No expansion loops or offsets are used. Similar to the continuous railroad construction, the pipeline requires heavy anchors or equivalent anchors placed at ends to control the end movements, and sufficient lateral restraining efforts are provided to prevent lateral buckling in pipelines. Details on cross-country pipelines are discussed in a separate chapter.

3.9.1 Problems Associated With Excessive Flexibility

Excessive piping flexibility comes from excessive loops and offsets in the piping system. Externally, the piping with excessive loops is flimsy and prone to vibrate. It is weak in resisting occasional loads, such as wind and earthquake. Internally, these excessive pipes and bends generate excessive pressure drop and excessive bumps to the fluid flowing inside. Aside from the increased operating cost due to the extra power required, pressure drop can cause vaporization of the fluid operating at near-saturate temperature. The near-saturate condition occurs at all condensate lines and bottom fluid lines from distillation towers, and so forth. When vaporization occurs at the portion of piping away from the pumps, it generates shaking forces due to slugs formed by the vapor. Once slug formation occurs, the whole piping system bangs and shakes violently, even if the system is not unsafe to operate. More discussions on slug flow are given in the chapter dealing with dynamic analysis.

The biggest problem associated with excessive flexibility occurs at pumping circuits. The excessive piping loops in this case are add-ons in an attempt to reduce piping loads on the pump. These loops are not in the scope of original fluid flow planning. Therefore, the added pressure drop is often not accounted for in the original system design. To ensure a smooth and efficient pump operation, a certain minimum amount of net positive suction head (NPSH) must be maintained at the pump inlet. Otherwise, vaporization and cavitation might occur at the suction side. This can substantially reduce the pump capacity and pump efficiency. The vapor eventually collapses when pressure inside the pump reaches above the saturation point. This collapse of vapor generates a huge impact force that can damage the internals of the pump.

Many pumps, such as condensate and boiler feed water pumps, are operating at near-saturate temperature. It is already very difficult to provide enough NPSH for them without the unexpected addition of flow resistance. Very often, storage tanks on the suction side have to be elevated to the top portion of the building to provide this required NPSH. In the layout of a petrochemical plant, the engineers generally do not have the luxury of elevating a storage tank as high as needed. Because the storage tank is often the bottom of a distillation or some other tower or vessel, to raise it means elevating the whole tower together with all other piping connected to it. It might also involve raising other towers as they are normally related by a fixed elevation difference. It is indeed very complicated even to raise the elevation by just a foot or two. Because of the lack of awareness regarding the need for loop, or due to the difficulty in foreseeing the size of the loop required, the extra pressure drop from the added loops is seldom accounted for in the original flow design. This naturally generates all types of operational problems for the pump. More discussions are given in Chapter 9.

3.10 FIELD PROVEN SYSTEMS

There are many installations that cannot be accurately analyzed even with current technology. It is either due to the shortfall of the technology or due to economic impracticality. The authors of the piping code had the vision to stipulate that “No formal analysis of adequate flexibility is required for a piping system which duplicates, or replaces without significant change, a system operating with a successful record;...” (ASME B31.3, par. 319.4.1; also B31.1, par. 119.7.1).

However, many modern-day engineers do not appreciate this foresight of the code. Not so many years ago, when it was customary to construct a plastic model of the plant for review by all associated parties, the engineers from an operating plant could easily spot a poorly designed system. They would say that they had never seen one like that, or that some other systems should be used because they have already proved themselves in the field. However, these comments were often not appreciated by computer-oriented engineers who would have insisted that the computer analyses had concluded otherwise.

The computer is mathematically correct in interpreting the data supplied. In piping flexibility, some systems in the field may not be readily analyzed by the computer due to a lack of accurate information. These systems include some low-temperature pump piping systems, short equipment connecting pipes, and others that involve small amounts of expansions or displacements. As discussed in Section 3.1, the analysis results of these systems depend entirely on the accurate stiffness of the end connections. If these systems were analyzed through a computer using the idealized data, they would have evolved to a design with excessive loops that may hamper the proper operation of the system.

This is not to say that the computer is not useful, but rather to emphasize that only accurate data will produce meaningful results. For systems involving small expansion or displacement, the exact stiffness of the boundary condition is very important. An analysis assuming an anchor here and another anchor there will not do the job correctly. In these cases, past experience is of the utmost importance.

Some words of caution are in order, however. The duplication of successfully operating system is only valid for systems do not involve time-dependent failure modes, such as creep or medium cycle fatigue. For a system with a time-dependent life cycle, it takes years to judge whether it is successful or not.

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